AquaCoach

Human-Centered Adaptive Hydration Tracking

Personalized goals, intelligent reminders, and privacy-first insights that help people build healthier hydration habits.

Student Research Poster

Developer: Aayush Keshari | **Program:** Computer Science
Advisors: Aishwarya Keshari, Shivani Gupta
Project Focus: Human-Centered Health Technology
Repository: github.com/aayushkeshari/AquaCoach

Problem & Goals

Many hydration tools rely on static goals and generic reminders. AquaCoach adapts hydration support to the person using it.

- Personalize daily hydration using weight, activity level, and weather.
- Reduce reminder fatigue with adaptive nudges and simple progress feedback.
- Make hydration tracking fast, intuitive, and accessible for everyday users.

Designed for Real Users

Target users were chosen from common hydration pain points: busy routines, inconsistent habits, and changing physical demands.

Busy Student

Needs quick logging and reminders that fit around classes, work, and variable schedules.

Active User / Athlete

Needs goals that respond to activity, temperature, and fluid loss during exercise.

Health-Conscious User

Needs clear insights, privacy, and motivation to build healthier daily habits.

Key Features

- 40+ beverage options with fast intake logging.
- Personalized hydration goal calculation.
- Adaptive reminders and streak-oriented feedback.
- Weather-aware and Apple HealthKit-ready design.
- Progress dashboard with trends and daily completion.

Human Impact

- Supports everyday well-being by encouraging consistent hydration habits.
- Uses simple interaction patterns to reduce friction and cognitive load.
- Designed to be inclusive across lifestyles, routines, and climates.
- Pairs technical innovation with privacy, trust, and usability.

Human-centered design decisions included intuitive flows, readable hierarchy, minimal jargon, and configurable reminders to respect user attention.

System Overview & Design

AquaCoach uses a modular architecture so hydration logging, personalization, reminders, and data storage can evolve without breaking the user experience.

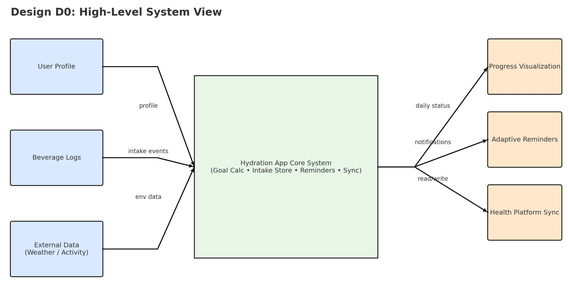


Figure 1. Level 0 system overview showing inputs, core logic, and outputs.

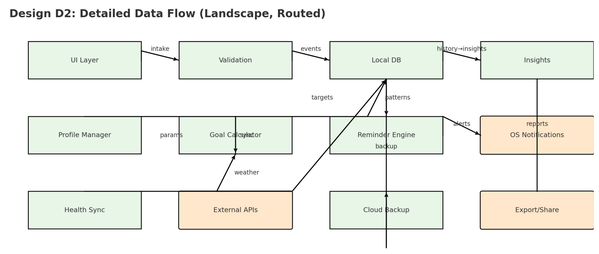


Figure 2. Detailed data flow for profile setup, goal calculation, storage, reminders, and export.

Design Specifications & Technologies

Functional requirements include beverage logging, personalized goal calculation, reminders, dashboard insights, and secure sync readiness.

Technologies

- SwiftUI for iOS interface development.
- SQLite / local storage for persistence.
- Apple HealthKit and UserNotifications integration.
- Weather API for environmental context.
- GitHub for version control and documentation.

Milestones & Deliverables

The project plan moves from architecture and prototyping to module development, integration, testing, and Expo demonstration.

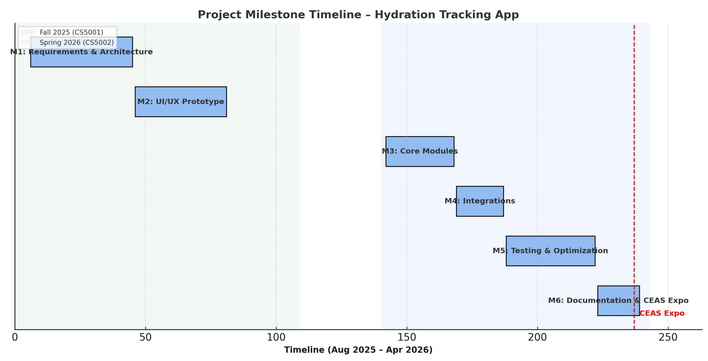


Figure 3. Milestone timeline from Fall 2025 architecture through Spring 2026 testing and Expo delivery.

Key deliverables: architecture diagrams, prototype UI, core modules, integrations, test plan, and CEAS Expo demo.

Progress & Results

- Completed requirements, design diagrams, and UI/UX wireframes.
- Established project workflow, repository structure, and documentation.
- Built the foundation for intake logging, goal calculation, and reminders.
- Drafted a 15-case test plan covering unit, integration, and performance scenarios.

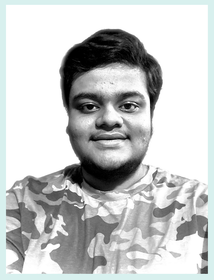
Early validation confirms persistence workflows, permission handling, and reminder logic under simulated usage conditions.

Challenges & Design Decisions

- Balancing personalized recommendations with a simple user experience.
- Preventing notification fatigue while still reinforcing healthy habits.
- Handling privacy-sensitive health data with secure, permission-based access.
- Managing architecture, documentation, and implementation as a solo developer.

Each challenge directly influenced the design: fewer taps, clearer feedback, stronger privacy defaults, and milestone-driven development.

Developer & Advisors



Aayush Keshari
Computer Science

My role spans architecture, prototyping, module development, integration, testing, and presentation. The project emphasizes clear technical communication alongside user-centered design.

Industry Advisors

Aishwarya Keshari
Shivani Gupta

Project contact
kesharah@mail.uc.edu
github.com/aayushkeshari/AquaCoach

Expo Demonstration

- Set up a user profile and generate a personalized hydration target.
- Log beverages and update progress in real time.
- Trigger reminder behavior and display dashboard insights.
- Show how the system scales toward full HealthKit-enabled deployment.

AquaCoach demonstrates how human-centered design, adaptive algorithms, and privacy-aware mobile engineering can improve everyday health behavior.
CEAS Expo 2026 | University of Cincinnati | Human-Centered Research & Design

GitHub: github.com/aayushkeshari/AquaCoach
Email: kesharah@mail.uc.edu